



System Integration and **Architecture**

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Objective



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- ☐ Define system integration and system architecture
- ☐ Explain why integration and architecture are important
- ☐ Identify real-world applications of integrated systems
- ☐ Differentiate standalone and integrated systems

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What is System Integration?

System integration is the process of **connecting different subsystems, applications, or components** so they function together as **one unified system**.

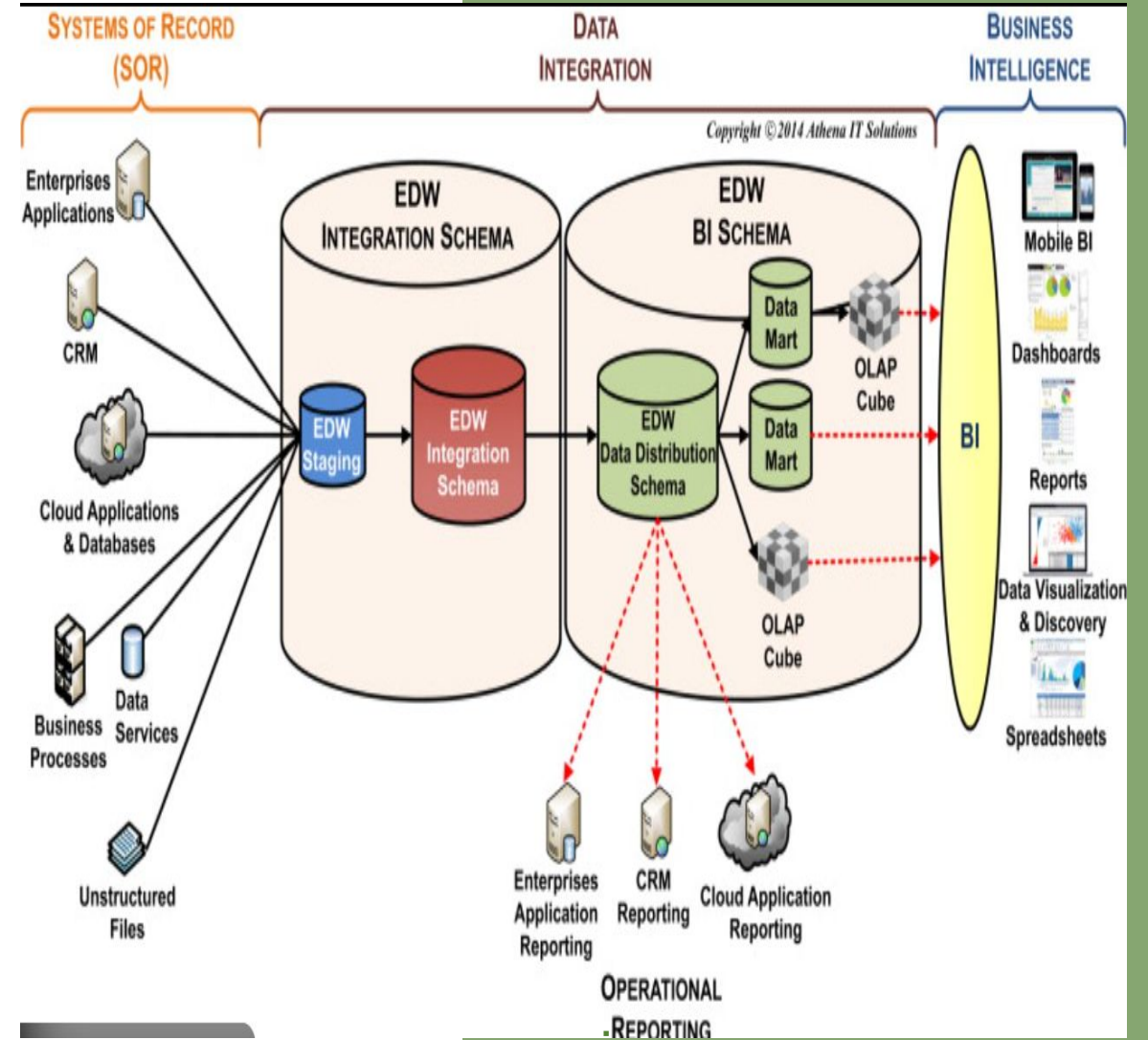
Instead of working in isolation, integrated systems:

- Share data
- Coordinate processes
- Support unified decision-making

Why System Integration Matters

Organizations rarely use a single system. Most rely on:

- Legacy systems
- New web-based applications
- Cloud platforms
- Third-party services



What is System Integration?



- ☐ Process of connecting multiple subsystems into one unified system
- ☐ Enables communication and data sharing among systems
- ☐ Supports automation and efficiency

Components of an Information System

1. HARDWARE

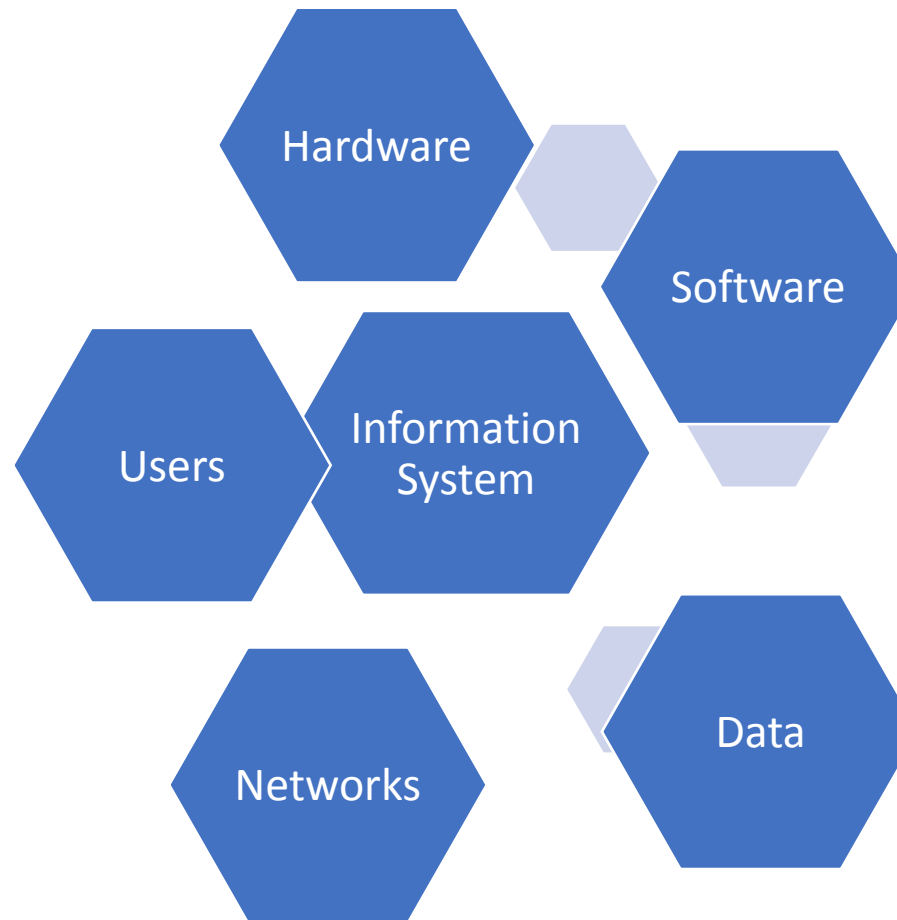
- Physical devices
- Ex: computers, servers, printers

2. SOFTWARE

- Program and application
- Ex: enrollment system, payroll software

3. DATA

- Raw facts and figures
- Ex: names, grades, attendance records



4. NETWORK

- System that connects computer and devices together to communicate and share data.
- Ex: school computer laboratory network (LAN)

5. USERS

- users, operators, managers, decision-makers
- Ex: students, teachers, clerks, administrators.

What is System Integration?



Key Types of Integration

-  **Data Integration** – sharing databases or data feeds
-  **Application Integration** – connecting software applications
-  **Process Integration** – aligning workflows across systems

Types of System Integration

- Vertical Integration

- Horizontal Integration
- Enterprise Application Integration (EAI)
- Data Integration
- Also called **Silo Integration**) is a type of system integration where systems are connected based on specific functional areas or departments, forming independent vertical “stacks.”



Advantages

- Easy to manage within departments
- Clear functional boundaries



Disadvantages

- Limited data sharing
- Creates data silos
- Not enterprise-wide



Types of System Integration

- Horizontal Integration
 - Uses a **common communication bus** (middleware) to integrate systems.
 - Systems communicate through a standardized platform

✓ Advantages

- Highly scalable
- Flexible
- Standardized communication
- Easier to add new systems

✗ Disadvantages

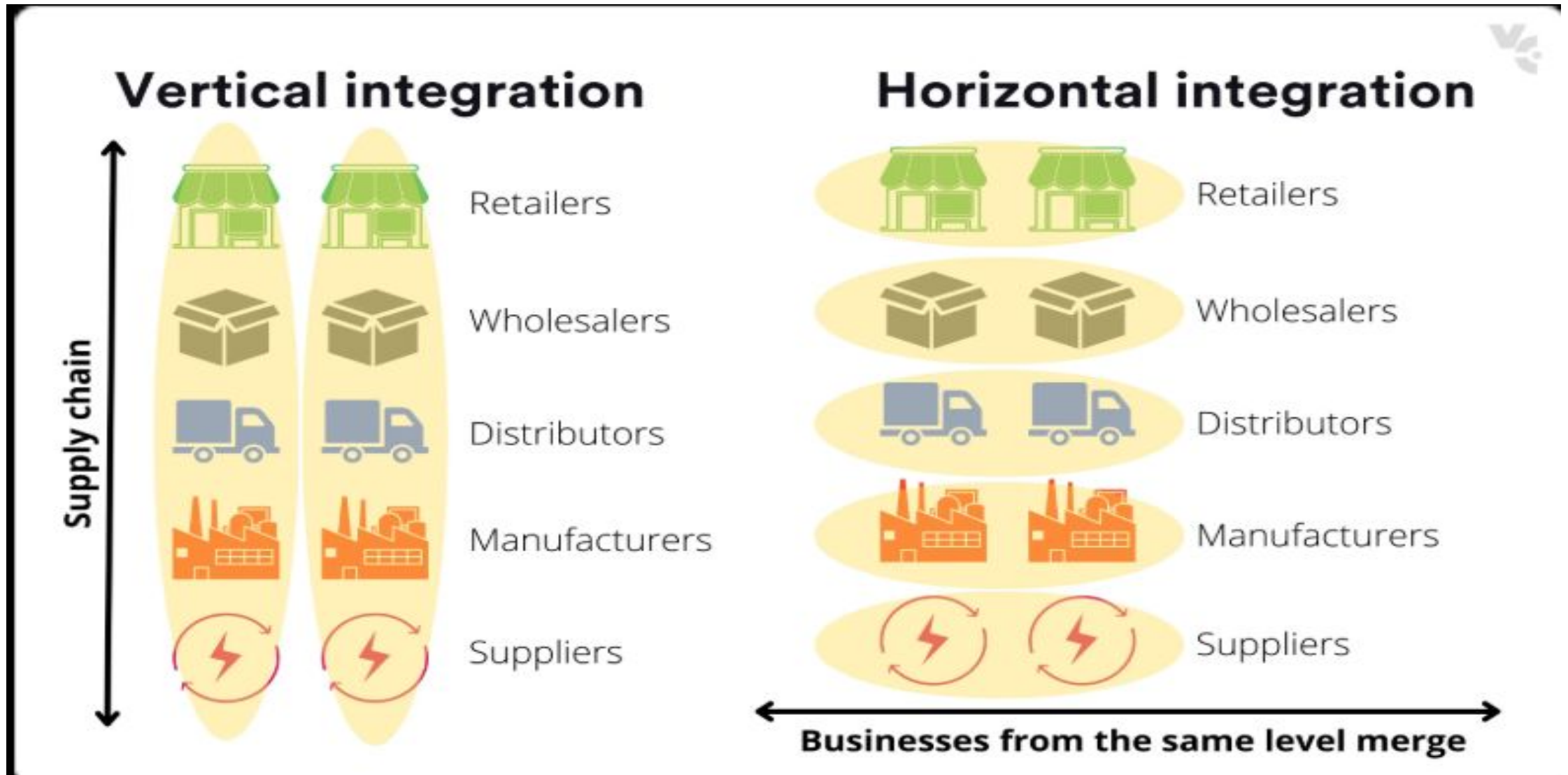
- Higher initial setup cost
- Requires strong architecture design

🎯 Best For

- Large organizations
- Government or defense systems
- Enterprise digital transformation



Types of System Integration



Common Integration Architecture

- Point-to-Point Integration

- Service-Oriented Architecture (SOA)
- Microservices Architecture
- Event-Driven Architecture

Point-to-Point (P2P) Integration is a system integration approach where each system is directly connected to another system using a dedicated connection.

There is **no central hub or middleware**.

Each pair of systems communicates independently.



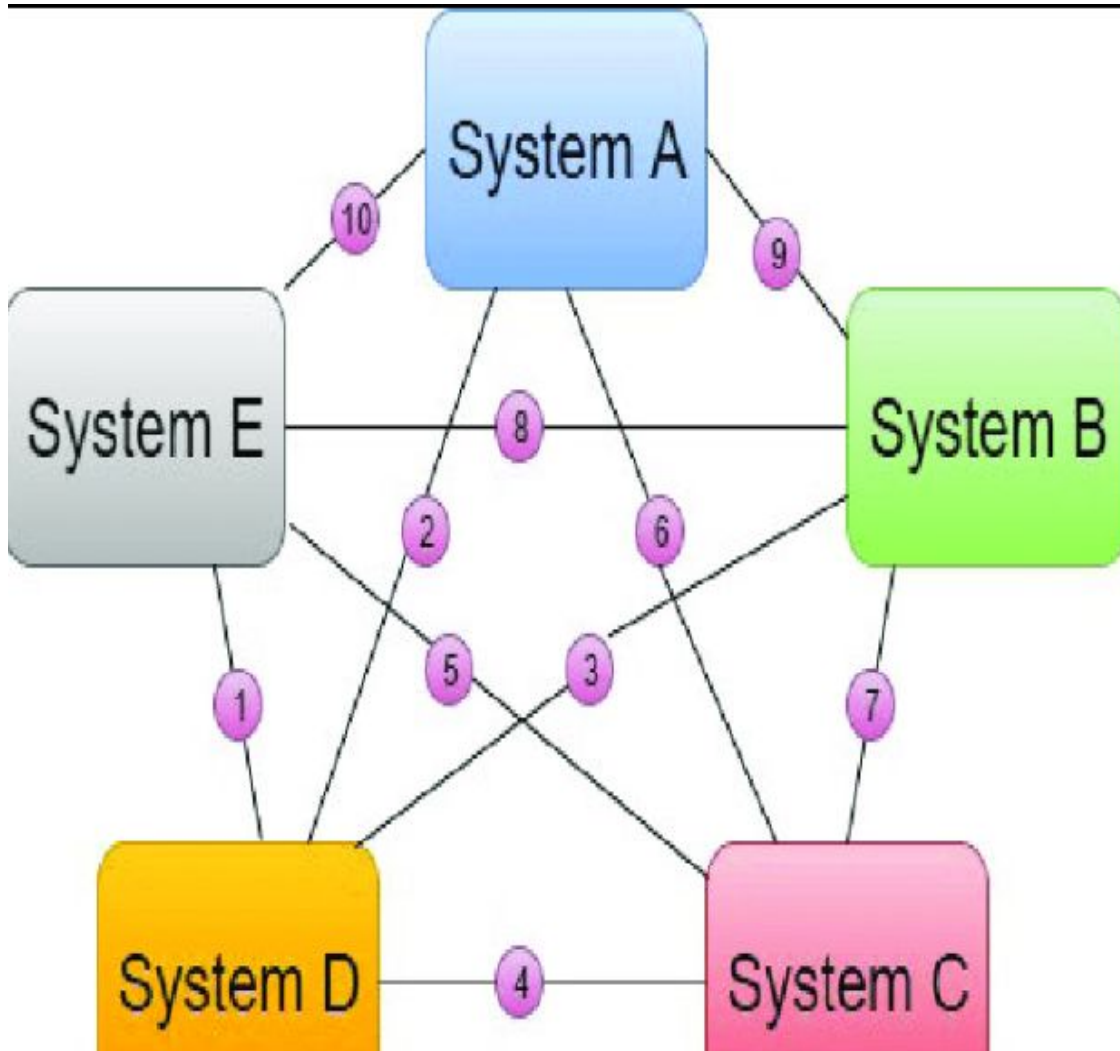
Advantages of Point-to-Point Integration

1. Simple for 2–3 systems
2. Quick to implement
3. Low initial cost
4. No need for middleware



Disadvantages

1. Hard to scale
2. Difficult maintenance
3. Complex troubleshooting
4. High long-term cost
5. Increased security risks
6. Tight coupling between systems



If we have 4 systems:

- System A connects to System B
- System A connects to System C
- System B connects to System D
- System C connects to System D

Every new system requires new direct connections.

As systems increase, connections grow rapidly

| | | |---|---| | 1 | 2 | | 3 | 4 | Why It Becomes Complex

If you have:

- 2 systems → 1 connection
- 3 systems → 3 connections
- 4 systems → 6 connections
- 5 systems → 10 connections

The number of connections grows exponentially.

Benefits of System Integration

- Improved data consistency and accuracy
 - Increased operational efficiency
 - Better decision-making through unified information
 - Reduced redundancy and costs

Challenges in System Integration

- Legacy systems and compatibility issues
 - Security and data privacy concerns
 - Complexity and high implementation cost

Role of System Architecture in Enterprise Systems,

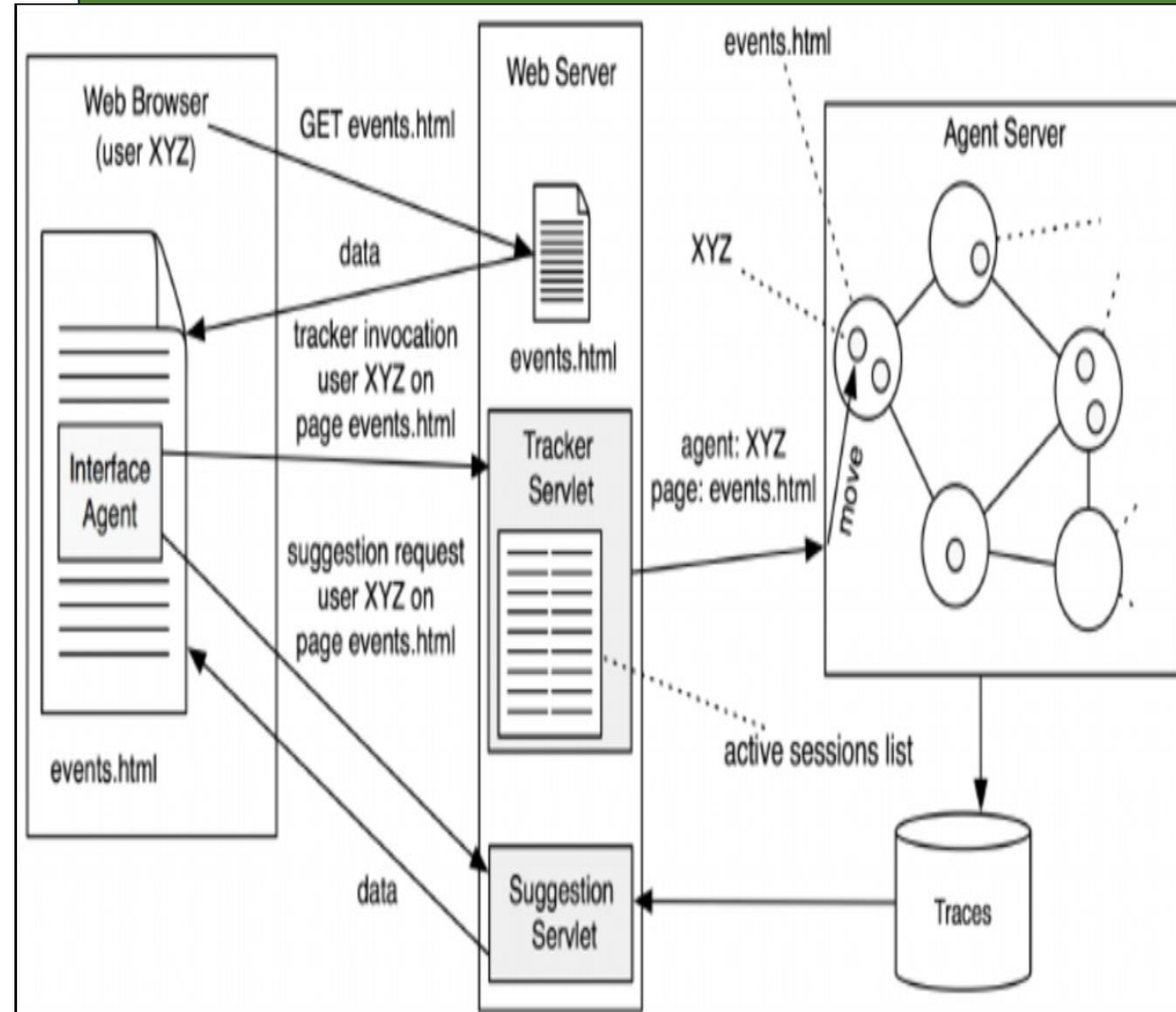
System architecture defines the:

- **Structure** of the system
- **Components** that make it up
- **Interfaces** between components
- **Interactions** on how components work together

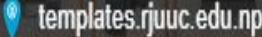
👉 It serves as the **blueprint** that guides how a system is designed, built, and integrated.

Think of it as the **blueprint** of the system:

- ❑ What components exist
- ❑ How they communicate
- ❑ Where integration happens



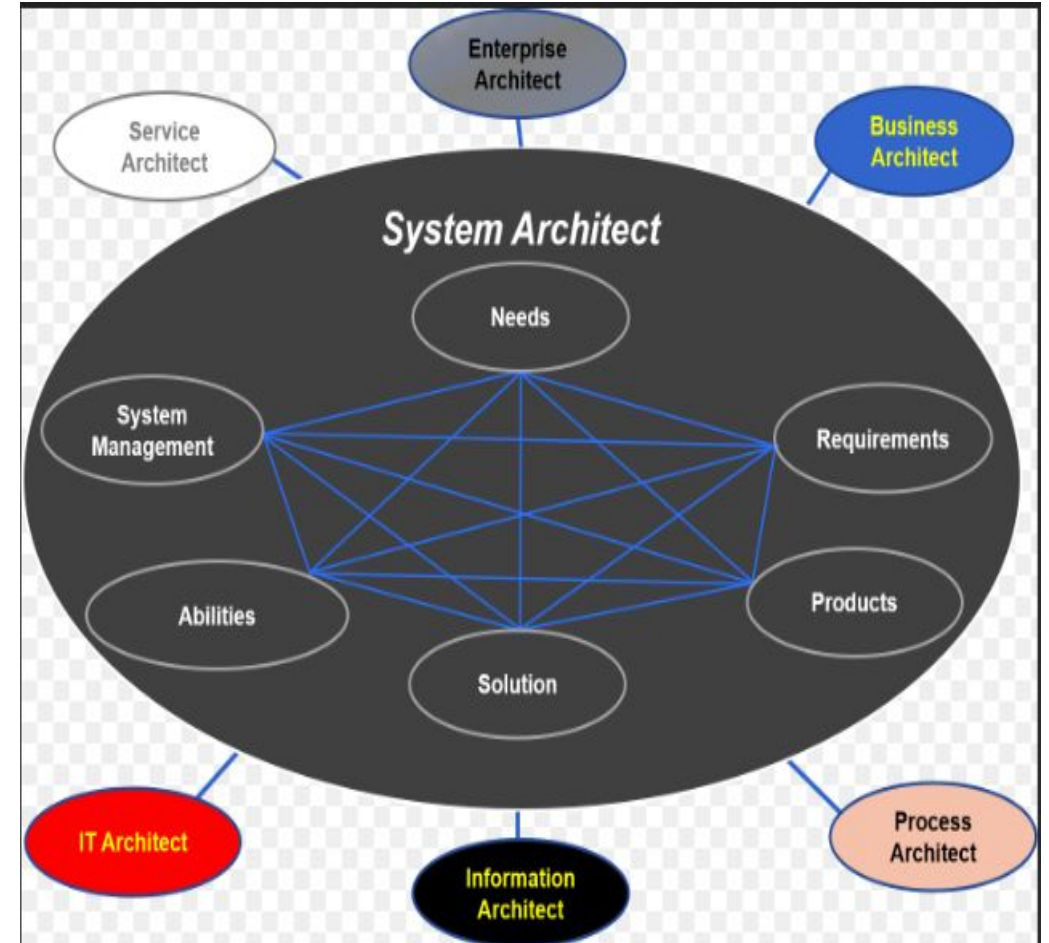
System: Architecture



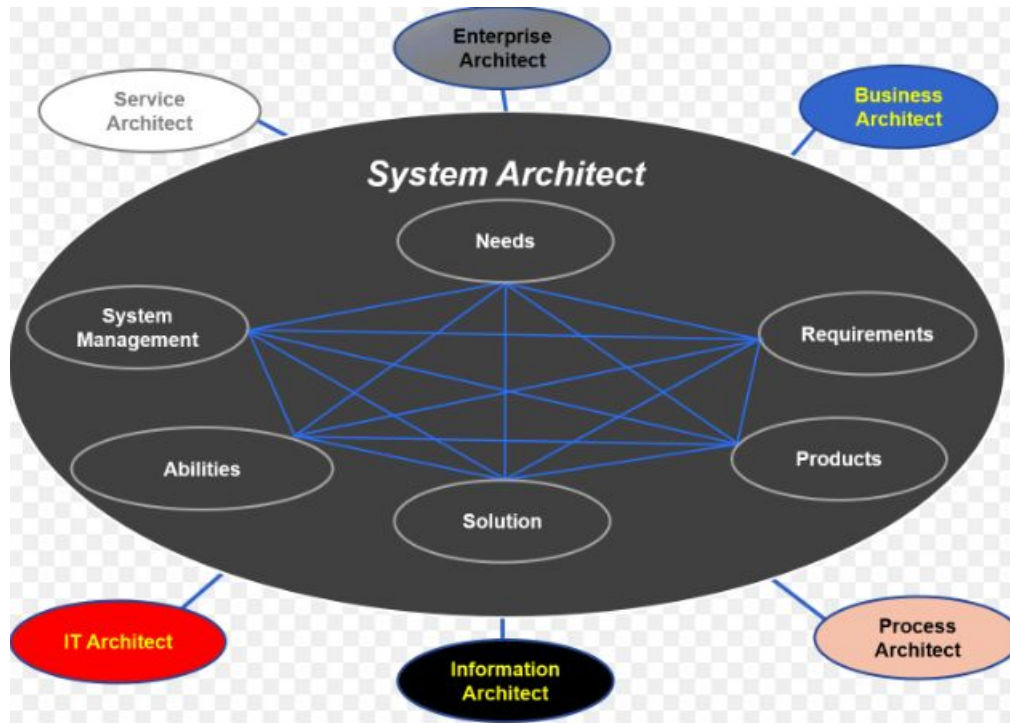
- ❑ Overall structure and design of a system
- ❑ Defines components and their interactions
- ❑ Acts as the system blueprint

Role of System Architecture

- ❑ Guides system development
- ❑ Ensures scalability and performance
- ❑ Supports future expansion



Importance of System Integration and Architecture



- Improves business efficiency and automation
- Ensures data consistency and accuracy
- Supports scalability and flexibility
- Reduces long-term costs

Real-World Application

- Enterprise systems (ERP, CRM)
- E-commerce platforms
- Banking and financial systems
- Healthcare information systems
- Government and academic systems



Class Activity

Activity:

- Identify one system you use daily
- List the subsystems it is connected to
- Explain why integration is important

Challenges in Integrating Heterogeneous Systems

How Architecture Helps Address These Challenges

🔌 Standard Interfaces (APIs)

- Enables consistent system communication
- Reduces dependency on specific technologies

🔗 Middleware and integration platforms

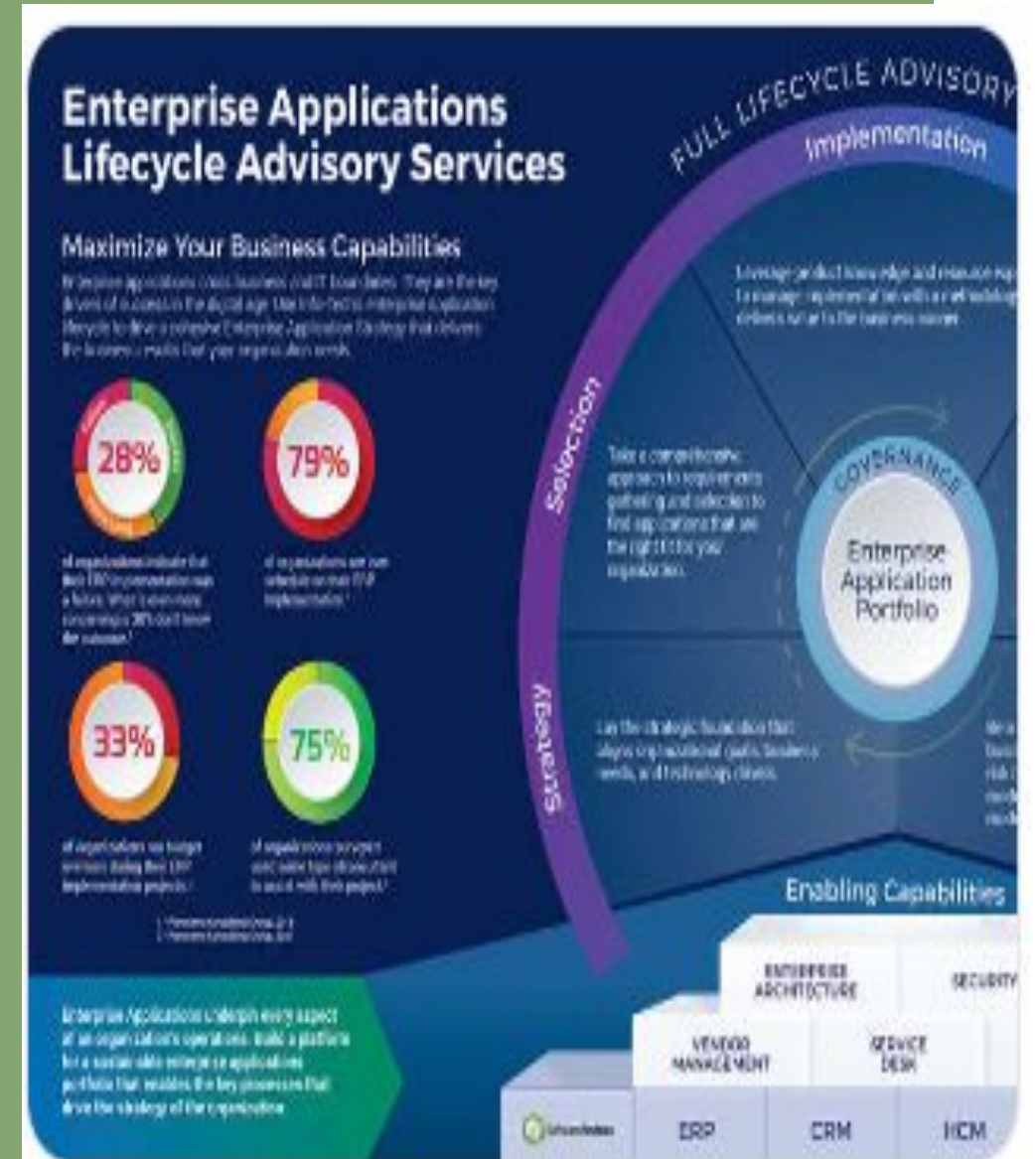
- Acts as a bridge between systems
- Handles data transformation and messaging

🧱 Modular design

- Breaks systems into manageable components
- Easier upgrades without affecting the whole system

📄 Clear documentation and governance

- Defines standards and responsibilities
- Ensures consistency, security, and control



Thank You

LIFE WITH STYLE